

UNIT 4.INTRODUCTION TO RASPBERRY PI & ARDUIANO

4.1 INTRODUCTION ON IOT DEVICES

IoT devices are the nonstandard computing devices that connect wirelessly to a network and have the ability to transmit data, such as the many devices on the internet of things (IoT).

IoT involves extending internet connectivity beyond standard devices, such as desktops, laptops, smartphones and tablets, to any range of traditionally "dumb" or non-internet-enabled physical devices and everyday objects. Embedded with technology, these devices can communicate and interact over the internet. They can also be remotely monitored and controlled.

4.2 BASIC BUILDING BLOCKS OF AN IOT DEVICE (BUILDING BLOCKS OF IOT)

Four things form basic building blocks of the IoT system –sensors, processors, gateways, applications. Each of these nodes has to have its own characteristics in order to form an useful IoT system.

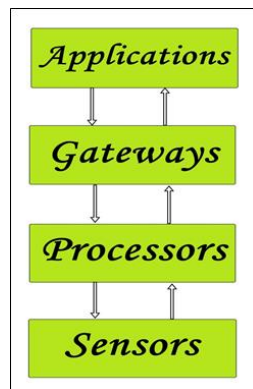


Figure 1: Simplified block diagram of the basic building blocks of the IoT

Sensors:

- These form the front end of the IoT devices. These are the so-called “Things” of the system. Their main purpose is to collect data from its surroundings (sensors) or give out data to its surrounding (actuators).
- These have to be uniquely identifiable devices with a unique IP address so that they can be easily identifiable over a large network.
- These have to be active in nature which means that they should be able to collect real-time data. These can either work on their own (autonomous in nature) or can be made to work by the user depending on their needs (user-controlled).
- Examples of sensors are gas sensor, water quality sensor, moisture sensor, etc.

Processors:

- Processors are the brain of the IoT system. Their main function is to process the data captured by the sensors and process them so as to extract the valuable data from the enormous amount of raw data collected. In a word, we can say that it gives intelligence to the data.
- Processors mostly work on real-time basis and can be easily controlled by applications. These are also responsible for securing the data – that is performing encryption and decryption of data.
- Embedded hardware devices, microcontroller, etc are the ones that process the data because they have processors attached to it.

Gateways:

- Gateways are responsible for routing the processed data and send it to proper locations for its (data) proper utilization.
- In other words, we can say that gateway helps in to and fro communication of the data. It provides network connectivity to the data. Network connectivity is essential for any IoT system to communicate.
- LAN, WAN, PAN, etc are examples of network gateways.

Applications:

- Applications form another end of an IoT system. Applications are essential for proper utilization of all the data collected.
- These cloud-based applications which are responsible for rendering the effective meaning to the data collected. Applications are controlled by users and are a delivery point of particular services.
- Examples of applications are home automation apps, security systems, industrial control hub, etc.
- In Figure 2, the extreme right block forms the application end of the IoT system.

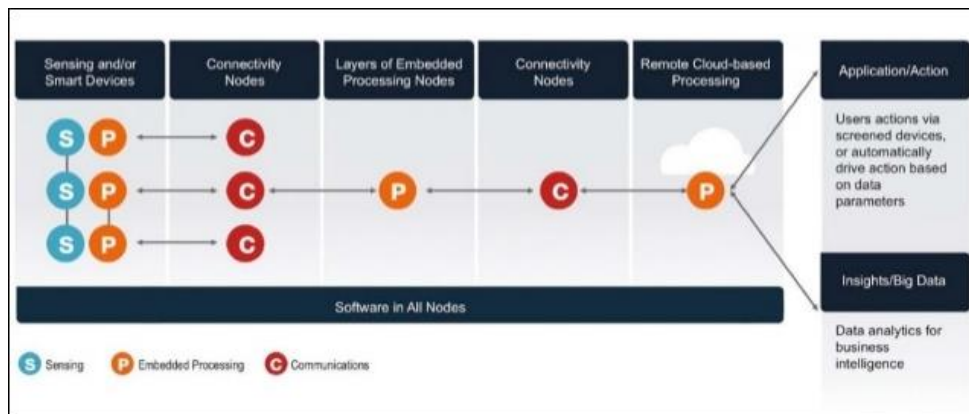


Figure 2: Basic building blocks of IoT

- In a nutshell, from the figure we can determine that the information gathered by the sensing node (end node) is processed first then via connectivity it reaches the embedded processing nodes that can be any embedded hardware devices and are processed there as well. It then passes through the connectivity nodes again and reaches the remote cloud-based processing that can be any software and is sent to the

application node for the proper applied usage of the data collected and also for data analysis via big data.

4.3 INTRODUCTION TO RASPBERRY PI (CONCEPTS, PURPOSE, APPLICATION AREAS)

Raspberry pi is defined as a minicomputer the size of a credit card that is interoperable with any input and output hardware device like a monitor, a television, a mouse, or a keyboard – effectively converting the set-up into a full-fledged pc at a low cost.

APPLICATION

1. DESKTOP PC

- Using Raspberry Pi, the microSD card, and a power supply, a simple desktop can be made. We would also need an HDMI cable and a suitable display, maybe an old monitor. A USB keyboard and mouse are also needed.
- The new version, which is Raspberry Pi 3, has built-in Wi-Fi and Bluetooth too. If a different model is used, compatible USB dongles would be required.
- Once everything is set up, and preferred operating system installed (the latest version of Raspbian), your desktop computer is ready to be used.

2. WIRELESS PRINT SERVER

- This requires installing Samba file-sharing software and CUPS (Common Unix Printing System). CUPS provide drivers for the printer and administration console.
- After this, Pi configuration is needed to ensure a Windows or Mac computers can access the printer via a network. The printer must have a USB cable.

3. MEDIA USAGE

- Many estimates suggest one of the main uses of Raspberry Pi is a Kodi media center. Several Kodi builds have been released as disk images. OSMC and OpenElec are among the most popular.
- Installing Kodi comes with some caveats. It is recommended that we install only safe and legal add-ons from the official Kodi repositories. Also, a Raspberry Pi running Kodi is vulnerable to a few security issues. Hence, setting up a VPN to encrypt data is recommended.

4. GAME SERVERS

- Raspbian, the default OS of pi comes with a special version of Minecraft game pre-installed. But, the applications of Raspberry Pi can be used as a game server as well. It is an excellent game server for Minecraft. If multiple Raspberry Pis are used, making one as a dedicated server, a great gaming experience can be achieved.
- Other multiplayer network games can be set up on the Raspberry Pi.

5. RETRO GAMING MACHINE

- Raspberry Pi is ideal as a retro gaming machine. it fits as one of the lightest components of a machine. Particularly, it's a version, The Raspberry Pi Zero, that can fit into small spaces for gaming projects. There are two main options, Recalbox and RetroPie. Other platforms can be emulated too. Classic MS-DOS PC gaming and Commodore 64 can also be set-up and also many other popular 16-bit games consoles.

6. ROBOT CONTROLLER

- There are many robot-controller Raspberry Pi projects. There is a dedicated robotics package for Pi, duly powered with the device battery and used to communicate and control robots.
- for robots, Pi Zero W can only be used. Zero, a slim line version of the Raspberry Pi, has features of onboard wireless connectivity suitable for lightweight robots.
- It's quite lighter than the Model B+ boards of version 2 and 3 of pi, and the low profile ensures it can be placed in an efficient position without having a concern about USB ports.

7. STOP MOTION CAMERA

- Using Python and a suitable mount (standard tripod for clay- or toy-based) and the availability of a well-lit area Stop motion camera can be built. But, this is a time-consuming process. One needs a good amount of practice to get good results.

8. TIME-LAPSE CAMERA COMBININGGE

- The Raspberry Pi camera module and different script creates another use that captures movies. This can be achieved by taking single frames with a time delay. Also needed is, perhaps a portable battery solution, and a tripod can be used. A smartphone tripod is most preferred to ensure the device remains sturdy.

9. FM RADIO STATION

Raspberry Pi can also be used to broadcast on FM radio. Pi can broadcast only over a short-range. A portable battery and soldering skills may be required here. Any audio which needs to broadcast will need to be loaded beforehand to the microSD card.

10. WEB SERVERS

- Another great application of Raspberry Pi are to create a web server out of it. What this means is that it can be configured to host a website much like any other server. It can host blogs too. First of all, the right software needs to be installed and that is Apache and its dependent libraries. A full LAMP stack can also be installed with PHP, MySQL, and Apache too. Setting up FTP is also helpful.
- Once all these steps as mentioned are completed, HTML files can be saved into the /www/ directory, and the webserver is ready to be used. Specific web software like WordPress can also be used once the server setup is complete.

4.4 COMPONENTS OF RASPBERRY PI (FEATURES OF RASPBERRY PI)

1. Central Processing Unit (CPU)

Every computer has a Central Processing Unit, and so does the Raspberry Pi. It is the computer's brain and carries out instructions using logical and mathematical operations. Raspberry Pi makes use of the ARM11 series processor on its boards.

2. HDMI port

Raspberry Pi board has an HDMI or High Definition Multimedia Interface port that allows the device to have video options of the output from the computer displayed. An HDMI cable connects the Raspberry Pi to an HDTV. The supported versions include 1.3 and 1.3. It also comes with an RCA port for other display options.

3. Graphic Processing Unit (GPU)

This unit, GPU or Graphic Processing Unit, is another part of the Raspberry pi board. Its primary purpose is to hasten the speed of image calculations.

4. Memory (RAM)

Random Access Memory is a core part of a computer's processing system. It is where real-time information is stored for easy access. The initial Raspberry Pi had 256MB RAM. Over the years, developers gradually and significantly improved the size. Different Raspberry Pi models come with varying capacities. The model with the maximum capacity presently is the Raspberry Pi 4 with 8GB RAM space.

5. Ethernet port

The Ethernet port is a connectivity hardware feature available on B models of Raspberry Pi. The Ethernet port enables wired internet access to the minicomputer. Without it, software updates, web surfing, etc., would not be possible using the Raspberry Pi. The Ethernet port found on Raspberry computers uses the RJ45 Ethernet jack. With this component, Raspberry Pi can connect to routers and other devices.

6. SD card slot

Like most other regular computers, Raspberry Pi must have some sort of storage device. However, unlike conventional PCs, it does not come with a hard drive, nor does it come with a memory card. The Raspberry Pi board has a Secure Digital card or SD card slot where users must insert SD cards for the computer to function. The SD card functions like a hard

drive as it contains the operating system necessary for turning the system on. It also serves to store data.

7. General Purpose Input and Output (GPIO) pins

These are upward projecting pins in a cluster on one side of the board. The oldest models of the Raspberry Pi had 26 pins, but most have 40 GPIO pins. These pins are pretty sensitive and should be handled carefully. They are essential parts of the Raspberry Pi device as they add to its diverse applications. GPIO pins are used to interact with other electronic circuits. They can read and control the electric signals from other boards or devices based on how the user programs them.

8. LEDs

These are a group of five light-emitting diodes. They signal the user on the present status of the Raspberry Pi unit. Their function covers:

- PWR (Red): This functions solely to indicate power status. When the unit is on, it emits a red light and only goes off when the unit is switched off, or disconnected from the power source.
- ACT (Green): This flashes to indicate any form of SD card activity.
- LNK (Orange): LNK LED gives off an orange light to signify that active Ethernet connectivity has been established.
- 100 (Orange): This light comes on during Ethernet connection when the data speed reaches 100Mbps.
- FDX (Orange): FDX light also comes during Ethernet connection. It shows that the connection is a full-duplex.

9. USB ports

Universal service bus (USB) ports are a principal part of Raspberry Pi. They allow the computer to connect to a keyboard, mouse, hard drives, etc. The first model of Raspberry Pi had only two USB 2.0 ports. Subsequent models increased this number to four. Raspberry Pi 4 and Pi 400, much newer models, come with a mix of USB 2.0 and USB 3.0 ports.

10. Power source

Raspberry Pi has a power source connector that typically uses a 5V micro USB power cable. The amount of electricity any Raspberry Pi consumes depends on what it's used for and the number of peripheral hardware devices connected.

4.5 Introduction to Arduino (Concept, purpose and Application areas)

Arduino is an open-source electronics platform that is based on beginner-level hardware and software.

The hardware component of an Arduino board is a programmable circuit board that is also known as a **microcontroller**. A microcontroller is a small computer with a processor, memory, and other peripherals designed for embedded applications. The software component is called Arduino **IDE** (Integrated Development Environment) where developers write and upload the code onto the microcontroller, it will be further explained later in the article.

Arduino is mainly used to build electronic projects for everyone – electricians, tinkerers, hobbyists, makers, and beginners. You can feed an Arduino board a set of instructions for it to carry out certain tasks. It is able to read the inputs and turn them into an output, for example, it will read a finger on a button and turn on an LED.

Let us now look at some reasons why one would use Arduino!

WHY USE ARDUINO ?

SIMPLICITY

Besides the main microcontroller chip, a microcontroller will require many different parts to work. What Arduino did was take away all the essential components of a microcontroller and design it so that it is effortless to operate on a piece of Printed Circuit Board (PCB) — making it very welcoming to all beginners!

Furthermore, Arduino's easy-to-use IDE software for beginners is a more manageable program to learn as it uses a simplified version of C++ compared to other programming software. Because of this, Arduino is commonly cited as the pathway for everyone looking to learn about microcontrollers. With it being optimized for users of all levels, even advanced users are also taking advantage of the Arduino IDE!

COMMUNITY

The Arduino community is vast and active, with users and organizations from all over the world. There is extensive documentation readily available online. A wide variety of tutorials and projects are pre-coded for you to learn and build using Arduino, allowing non-coders to get started very easily.

CHEAP

[Arduino Micro](#) only costs \$18.90, but it packs a powerful ATmega32U4 and has 20 digital pins.

You can also get your very own Raspberry Pi – RP2040 microcontroller [Arduino Nano](#) for only \$25.50. It is filled with 20 digital pins and has WiFi and Bluetooth connectivity thanks to the Nina W102 uBlox module.

[Arduino Nano v3](#) only costs \$19.80 but is powered with the ATmega328 and boasts 14 digital pins with 32 kb of flash memory!

CROSS-PLATFORM

Arduino IDE is also cross-platform, which means you can run it on Windows, Mac OS, and Linux operating systems compared to other microcontrollers, which primarily run only Windows.

WIDE VARIETY

Arduino has many more variations for you to choose from, you can pick the one that suits your project needs the most! There is currently a plethora of Arduino-compatible boards on the market. It is all supported very strongly by the Arduino community. It ranges from thumb-sized boards (21×17.5mm) to mega boards (115x75mm), each serving different project needs.

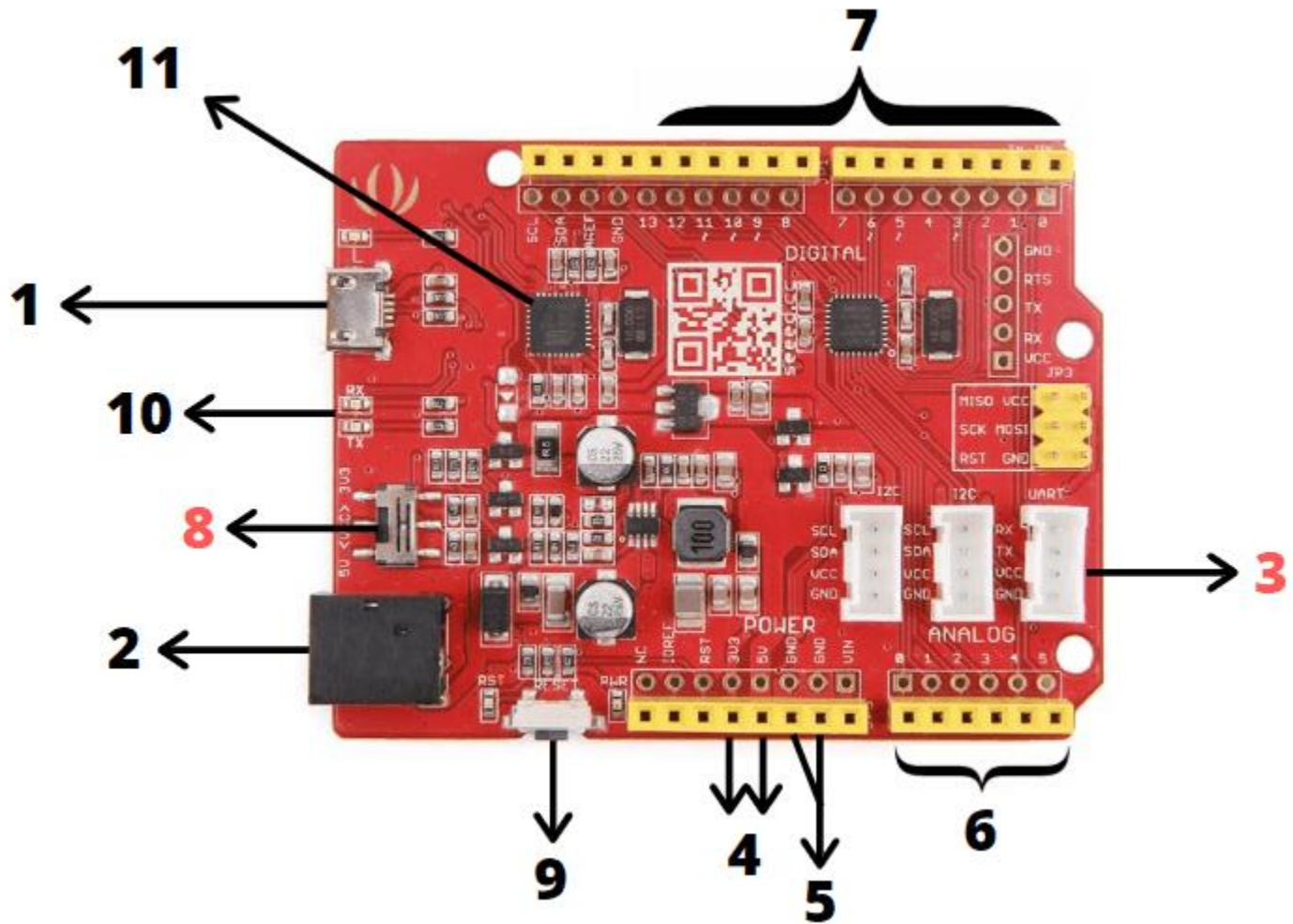
Introducing the [Seeeduino product line](#) and [XIAO series](#) by Seeed Studio. It caters to almost every project you can think of, ranging from small projects like wearables to mega IoT projects!

We will talk more about all the different types of Arduino boards later on. But now, we will take a look at what makes up an Arduino board.

WHAT MAKES UP AN ARDUINO BOARD?

The physical hardware of Arduino is the board itself. However, when it comes to Arduino boards, there are multiple varieties with different functionalities.

Today we will be looking at our [Seeeduino V4.2](#), which has the same functions as one of the most popular Arduino boards – the Arduino UNO.



Compared to the Arduino UNO, it has some extra features highlighted in red and can only be found on our Seesduino boards!

1 - USB Input

- USB Port connects the board to your PC for programming and powering up the Arduino board.
- This USB connection is important as it will be through this port that you will upload your code onto your Arduino board.

To learn more about how to Upload Code on your Arduino, you can check out our tutorial on [How to Upload Code on Arduino](#).

2 - DC Input

- The DC power jack allows your Arduino board to be powered from a wall adapter so that you can supply more power to your project if needed.

3 - Grove Connectors

- These Grove connectors can only be found on our Seesduino boards.

- Seeed Studio has a variety of sensors and devices that can use this I2C or UART connection.
- With our Grove Connectors, you can easily plug-in modules with Arduino without any soldering or jumper systems.
- We will elaborate more about Grove later in our Sensors and shields section.

4 – 3.3V and 5V Pins

- As the name specifies, the 3.3V and 5V pins supply volts of power to your modules. The 3.3V pin provides 3.3 volts of power, while the 5V pin provides 5 volts of power.

5 – GND pins

- The GND (Ground) pins are used to ground your circuit.
- GND means this pin is at zero voltage with respect to the power supply and ground plane of the circuit board.

6 – Analog Pins

- The analog pins allow the board to read signals from an analog sensor, like a light sensor and convert it into a digital value.
- Even though the primary function of the analog pins for most Arduino users is to read analog sensors, the analog pins also have all the functionality of general-purpose input/output (GPIO) pins.

7 – Digital Pins

- The digital pins on the Seeeduino or Arduino UNO are on Pin 0 to 13.
- They allow the board to read digital inputs like pushing a button and output something like turning on an LED.

8 – System Power Switch

- This system power switch can only be found on our Seeeduino boards.
- This slide switch is used to change the board's logic level and operating voltage to either 5V or 3.3V, which is helpful as if you want to save power, you can set it to 3.3V.

9 – Reset Button

- This reset button allows you to reset the board and restart any code uploaded on your Arduino board. Once pressed, the reset pin will be temporarily connected to the ground.
- This reset button is handy for your projects if your code does not repeat itself, but you wish to test it multiple times.
- This button is conveniently placed on the side to allow you to reset the Seeeduino board even when a shield is placed on top. This is different from other Arduino boards where the button is placed on top, making it hard to access.

10 – RX/TX Indicator

- Also known as the Transmit and Receive Indicators
- The TX and RX LED indicators are connected to the TX and RX of the USB-to-UART chip.
- They work automatically and let you know when the board is sending or receiving data. For example, when you upload a program onto your Arduino board.

11 – Microcontroller

- The Seeeduno V4.2 and Arduino UNO are based on the microcontroller ATmega328P.
- This is the primary chip that acts as your Arduino board's brain.
- They allow you to program your Arduino for it to be able to execute commands and decisions based on the code.
- You will have to know which type of microcontroller your board uses before loading a new program from the Arduino Software.
- Even though the microcontroller on the Arduino boards is different, their difference is not significant. The only difference you may notice is the different amounts of onboard memory.

4.6 Difference between Raspberry pi and Arduiano

S No.	Arduino	Raspberry Pi
1.	Control unit of Arduino is from Atmega family.	While control unit of Raspberry Pi is from ARM family.
2.	Arduino is based on a microcontroller.	While Raspberry Pi is based on a microprocessor.
3.	It is designed to control the electrical components connected to the circuit board in a system.	While Raspberry Pi computes data and produces valuable outputs, and controls components in a system based on the outcome of its computation.
4.	Arduino boards have a simple hardware and software structure.	While Raspberry Pi boards have a complex architecture of hardware and software.
5.	CPU architecture: 8 bit.	CPU architecture: 64 bit.

S No.	Arduino	Raspberry Pi
6.	It uses very less RAM, 2 kB.	While Raspberry Pi requires more RAM, 1 GB.
7.	It clocks a processing speed of 16 MHz.	While Raspberry Pi clocks a processing speed of 1.4 GHz.
8.	It is cheaper in cost.	While Raspberry Pi is expensive.
9.	It has a higher I/O current drive strength.	While Raspberry Pi has a lower I/O current drive strength.
10.	It consumes about 200 MW of power.	While it consumes about 700 MW of power.
11.	Its logic level is 5V.	Its logic level is 3V.
12.	It does not have internet support.	It has inbuilt Ethernet port and WiFi support.
13.	It has higher current drive strength.	It has lower current drive strength.
14.	Some of the applications of Arduino are traffic light countdown timer , Weighing machines , etc.	Some of the application of Raspberry Pi are Stop motion cameras , Robot Controllers , Game Servers.
15	operating system are required in arduino.	Operating System is required in reaspberry pi.

S No.	Arduino	Raspberry Pi
16	two tiny cores Arduino with 32 Mhz	Single core and 700 MHz